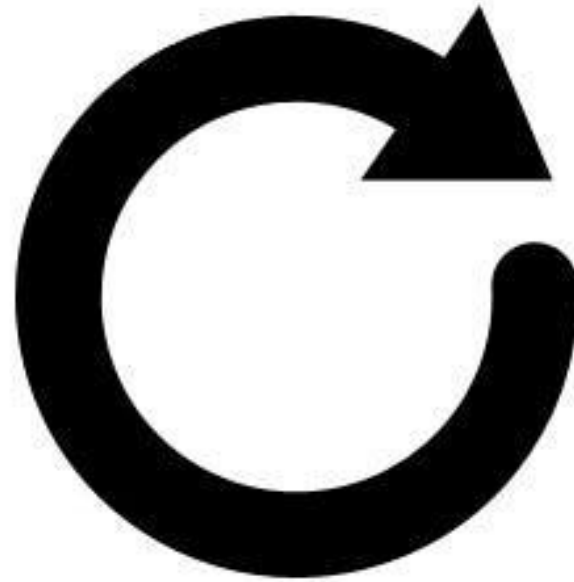


Loops

CSCI 1101 - Introduction to Computer Science

Looping Statements

- A loop statement allows us to execute a statement or group of statements multiple times.



LOOPS REPEAT
ACTIONS...
SO YOU DON'T HAVE TO

Types of Loops

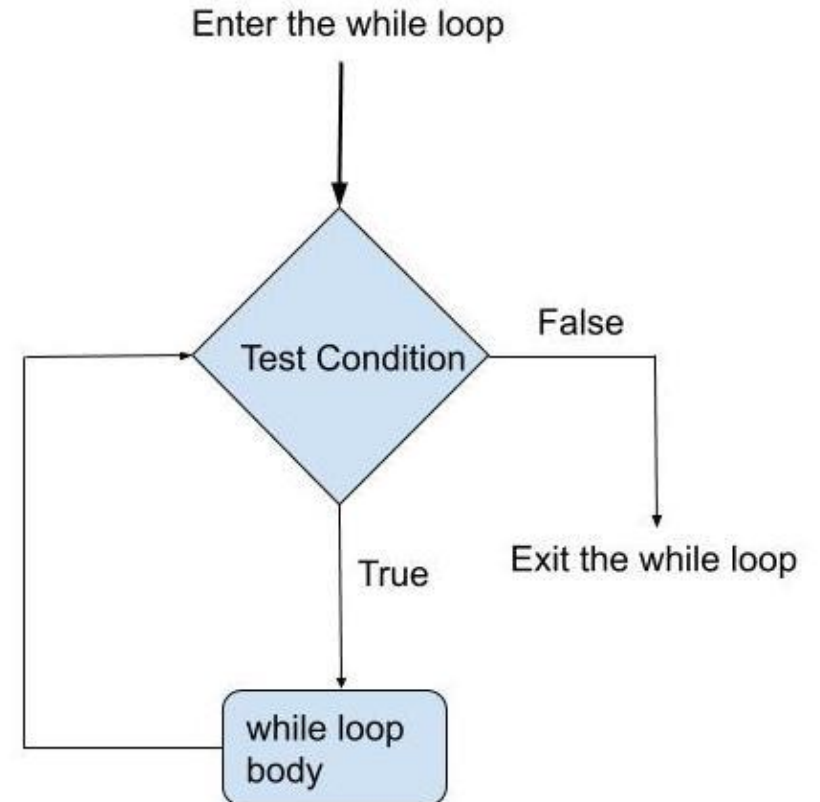
- **While Loop** – Repeats a statement or group of statements WHILE a given condition is **True**. It tests the condition before executing the loop body.
- **For Loop** – Repeats a statement or group of statements FOR a fixed number of times. It is frequently used to traverse the data structures like Lists, Tuples, or Dictionaries.

While Loop

Syntax

```
while ( condition ):  
    body of loop
```

Flow Chart



For our condition we use Boolean values or Boolean operators

Boolean Relational Operators

Operator	Definition
>	Greater Than
<	Less Than
>=	Greater Than or Equal To
<=	Less Than or Equal To
==	Equal To
!=	Not Equal To

Boolean Logical Operators

X	Y		X and Y	X or Y	not X
True	True		True	True	False
True	False		False	True	
False	True		False	True	True
False	False		False	False	

Infinite Loops

- Loops must contain within themselves a way to terminate
 - Something inside a while loop **must eventually make the condition false**
- **Infinite loop**: loop that does not have a way of stopping
 - Repeats until program is interrupted

Program to print 1 to 5

Example

Output

```
x = 1
```

```
while x < 6:
```

```
    print( x )
```

```
    x = x + 1
```

1

2

3

4

5

Program to determine which numbers are odd and even between 1 and 10

Example

```
x = 2
while x < 10:
    if x % 2 == 0:
        print( x, " is even" )
    else:
        print( x, " is odd" )
    x = x + 1
```

Output

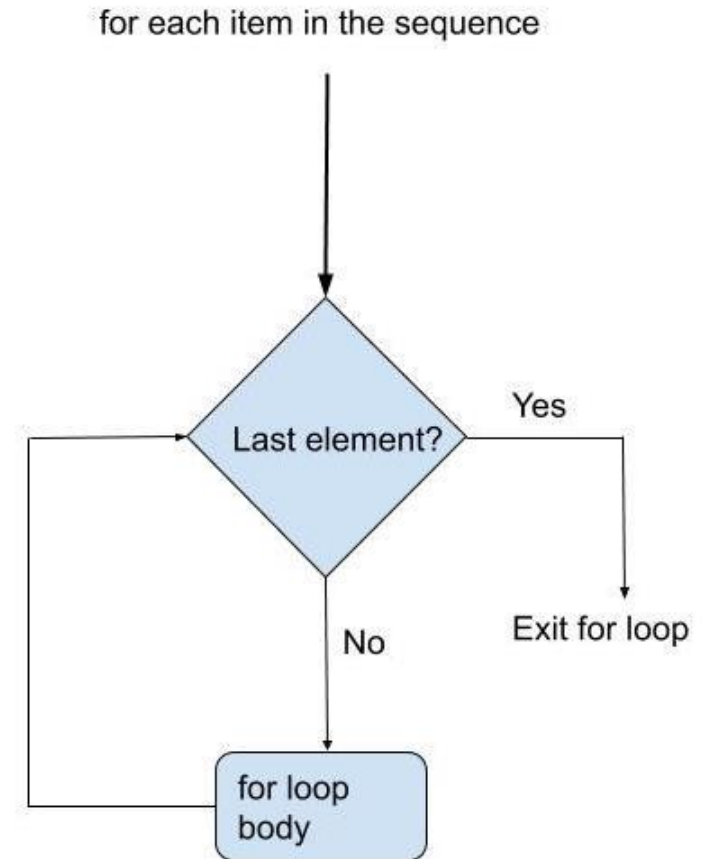
```
2 is even
3 is odd
4 is even
5 is odd
6 is even
7 is odd
8 is even
9 is odd
```


For Loop

Syntax

for *iterating_var* **in** *sequence*:
 body of loop

Flow Chart



Program to print 1 to 5

Example

```
for x in range(1, 6):  
    print( x )
```

Output

1

2

3

4

5

Program to print 5!
($5! = 1 \times 2 \times 3 \times 4 \times 5$)

Example

```
factorial = 1
for x in range(2, 6):
    factorial = factorial * x
print( "5! = ", factorial )
```

Output

5! = 120